

Homework 4

For Thursday, September 24, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. You may use the results covered in Lectures 1–5. When a proof directly from the definition is requested, use the epsilon–delta definition. All functions are real-valued. Limits and continuity are considered within the stated domain; at an included endpoint of an interval, continuity is one-sided.

1. Let $f : [0, \infty) \rightarrow \mathbb{R}$ satisfy $f(0) = 2$. Write the epsilon–delta definition of continuity of f at zero, and then write its negation. Include the restriction on x in both statements.

2. For $x \neq 2$, let

$$f(x) = \frac{x^2 - 4}{|x - 2|}.$$

Find the left-hand and right-hand limits at 2, and determine whether the two-sided limit exists.

3. For $k \in \mathbb{R}$, define F_k on $[-5, \infty)$ by

$$F_k(x) = \begin{cases} \frac{\sqrt{x+5}-3}{x-4}, & x \neq 4, \\ k, & x = 4. \end{cases}$$

Find the value of k for which F_k is continuous on its whole domain.

4. Find all $m, b \in \mathbb{R}$ for which the function

$$f(x) = \begin{cases} x^2, & x \leq 0, \\ mx + b, & 0 < x < 2, \\ \sqrt{x+7}, & x \geq 2 \end{cases}$$

is continuous on $\mathbb{R}$.

5. Find the real domain of

$$h(x) = \frac{\sqrt{4-x^2}}{\sqrt{x+2}-1}.$$

Prove that h is continuous at every point of its domain, explaining which one-sided convention applies at each included endpoint.

6. Let $f(x) = \sqrt[4]{1-x}$, with domain $(-\infty, 1]$. Prove directly from the definition that f is continuous at 1. Give an explicit δ for every $\varepsilon > 0$.
7. Suppose f and g are defined on an open interval I containing a , are continuous at a , and satisfy $f(a) < g(a)$. Prove that there is an $r > 0$ such that

$$x \in I, \quad |x - a| < r \quad \implies \quad f(x) < g(x).$$

8. Let f and g be continuous on an interval I . Prove that

$$H(x) = \max\{f(x), g(x)\}$$

is continuous on I . You may use the identity

$$\max\{u, v\} = \frac{u + v + \sqrt{(u - v)^2}}{2}.$$

9. Let $f(x) = x^2$ on $\mathbb{R}$. Construct a function $g : \mathbb{R} \rightarrow \mathbb{R}$ which is discontinuous at zero but for which $g \circ f$ is continuous at zero. Verify both properties.
10. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$, with $f(1) = 2$ and $g(2) = -1$. Suppose

$$\begin{array}{ll} |f(x) - 2| \leq 3|x - 1| & \text{when } |x - 1| < 1, \\ |g(y) + 1| \leq 2|y - 2| & \text{when } |y - 2| < 1. \end{array}$$

Prove directly that $g \circ f$ is continuous at 1, giving an explicit δ for every $\varepsilon > 0$.

11. Use the Intermediate Value Theorem to prove that

$$x^3 - 4x + 1 = 0$$

has at least three distinct real solutions. Give three disjoint open intervals with integer endpoints, each containing a solution. You do not need to solve the equation.

12. Let

$$D = [-2, -1] \cup [1, 2], \quad f : D \rightarrow \mathbb{R}, \quad f(x) = \frac{1}{x}.$$

Explain why f is continuous at every point of D and has $f(-2) < 0 < f(2)$, yet never equals zero. Identify the hypothesis of the Intermediate Value Theorem that is missing on $[-2, 2]$.

13. Suppose $f : [0, 1] \rightarrow [0, 1]$ is continuous. Prove that there is a $c \in [0, 1]$ for which $f(c) = c$. *Hint:* consider $f(x) - x$, including its values at both endpoints.
14. Let I be a nonempty interval and let $f : I \rightarrow \mathbb{R}$ be continuous. Suppose $f(x)^2 = 1$ for every $x \in I$. Prove that either $f(x) = 1$ for every $x \in I$, or $f(x) = -1$ for every $x \in I$.
15. Draw the graph of a continuous function $f : [0, 4] \rightarrow \mathbb{R}$ such that $f(0) = f(4) = 0$ and the equation $f(x) = 1$ has exactly three solutions. You may use line segments. Mark the three solutions clearly and explain why there are no others.